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Mitigation of antibiotic resistance: the efficiency of a hybrid subsurface flow constructed wetland in the removal of resistant bacteria in wastewater

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ABSTRACT

This research investigates the effectiveness of a lab-scale hybrid subsurface flow constructed wetland (HSSFCW) for removing wastewater contaminants, including antibiotic-resistant bacteria (ARB), genes (ARGs) and antibiotics. The results suggested that HSSFCW demonstrated a high removal efficiency for COD (89%) and BOD (88.9%), while lower efficiencies were observed for salts, TDS, EC, and TKN. Further, various bacteria such as *Enterobacter cloacae*, *Serratia liquefaciens* and *Serratia odorifera* were detected in the plant rhizosphere, while *Acinetobacter baumannii* and *Staphylococcus spp.* were identified as biofilm formers on the wetland media. The mean removal efficiency of 70.44, 65.99, 70.66 and 51.49% was observed for total heterotrophic bacteria; Cefixime (Cef)-, Ciprofloxacin (Cip)-, and Linezolid (Lzd)-resistant bacteria. Upon chlorination of effluent samples, Cef-, Cip- and Lzd-resistant bacteria were effectively inactivated at 30, 15 and 7.5 mg Cl₂ min/L, respectively. The wetland achieved a removal efficiency of 83.85% for Cip and 100% for Lzd at week 12 with $p=0.040$ and $p<0.001$, respectively. Further, a log reduction of 0.66 for *16S*, 0.82 for *blaTEM*, 0.61 for *blaCTX*, and 0.48 for *blaOXA* was observed. Thus, HSSFCW was observed to be efficient in removing organic contaminants, ARBs, ARGs and antibiotics from domestic wastewater and can be upgraded under natural environments.

NOVELTY STATEMENT

This study provides a novel contribution to the wastewater treatment processes by utilizing a hybrid constructed wetland for effectively removing antibiotic-resistant bacteria, including cefixime-, ciprofloxacin-, and linezolid-resistant strains, from domestic wastewater. Our results provide a comprehensive understanding of multi-target pollutant removal through reduction in multiple *bla* genes, resistant bacteria and antibiotics, such as ciprofloxacin and linezolid.

KEYWORDS

Antibiotic resistance; antibiotic resistant genes; constructed wetlands; environmental restoration and remediation; pollutant removal; wastewater treatment

Introduction

Antimicrobial resistance (AMR) is a worldwide threat to human, animal and environmental health. The problem of antibiotic resistance in developing countries is even more, where the burden of infectious diseases is higher, and access to primary healthcare is inadequate and expensive. In these countries the misuse, overuse and dumping of antibiotics is a common practice (Uluseker *et al.* 2021). This widespread resistance problem can lead to treatment failures and an increased mortality rate. The lack of proper wastewater management and removal of pollutants ultimately causes the contamination of rivers, lakes, ponds and groundwater. There are numerous reports indicating the widespread presence of antibiotic-resistant bacteria in different environments such as groundwater, surface water, treated water, drinking water, soil, and aquatic sediments. The most common source

of such contamination is the release of unregulated and untreated sewage from the residential areas (Dires *et al.* 2018). Municipal wastewater consisting of a consortium of microorganisms with abundant nutrients and antimicrobial agents such as antibiotics, becomes a favorable environment for the survival and dissemination of these pathogens (Sabri *et al.* 2021). The release of low concentrations of antibiotics in the environment exerts greater selective pressure on the bacterial communities in the soil and water (He *et al.* 2021). Even though antimicrobials have a limited half-life with a range of few hours to a couple of days, their decomposed products could drive the evolving mechanisms of AMR (Wang *et al.* 2020). The presence of antibiotic-resistant genes (ARGs) further complicates the issue of wastewater contamination (da Silva *et al.* 2022). The genetic nature of DNA and its ability to be transferred by mobile genetic elements facilitates the process of horizontal gene transfer between

bacterial species causing widespread dissemination of ARGs. While, ARGs are not biodegradable, their increased abundance in wastewater upon bacterial cell death can pose significant environmental implications (Zhao *et al.* 2023).

Despite the fact, conventional wastewater treatment plants (WWTPs) effectively reduce the organic load of wastewater, adequate removal of resistant bacterial population and the emerging contaminants such as antibiotics and ARGs are not significantly reduced (Rizzo *et al.* 2013; Topal and Topal 2015). Decentralized treatment technologies such as constructed wetlands are easy to install, have low operational cost and do not require regular maintenance. They are thus a better alternative for the treatment of wastewater compared to other conventional WWTPs in developing countries (Santos *et al.* 2019). Constructed wetlands can remove pathogenic, indicator, and ARBs more efficiently than other wastewater treatment systems, employing mechanisms such as adsorption, sedimentation, filtration, toxin-release from plants, biofilm-formation and natural die-off (Ma *et al.* 2020). The operating parameters of constructed wetlands that considerably influence the fate of ARGs and antibiotics include substrate media properties, species and density of plantation, temperatures, hydraulic flow rate and retention time. Moreover, studies determining the prevalence of antibiotics and ARGs concentrations in wastewater after treatment through constructed wetlands must be examined. Different configurations of constructed wetlands can be designed and built to enhance its treatment efficiency. One such modification of CWs such as the Hybrid CW arrangement consists of a two-stage design comprising parallel vertical and horizontal flow cells (De Lille *et al.* 2021). These cells are connected to each other and can be either composed of surface flow or subsurface flow beds. Further treatment including disinfection by chemical methods can also be beneficial in controlling the spread of antibiotic-resistant bacteria in WWTPs. One such tertiary treatment method of wastewater is achieved through chlorination, which has been proven effective in bacterial inactivation through microbial damage of DNA and proteins, thus impairing its ability to survive and multiply (Yuan *et al.* 2015). This treated wastewater can then be utilized for irrigation purposes or can be safely discharged into water bodies with minimal risk of pathogen contamination.

Although constructed wetlands have been extensively utilized for wastewater reclamation (Biswal and Balasubramanian 2022), removal of heavy metals (Topal and Topal 2020; Saeed *et al.* 2021), antibiotics (Topal *et al.* 2015), and antibiotic-resistant genes (Chen *et al.* 2022), still limited studies have focused on their effectiveness in removing both physicochemical pollutants and antibiotic resistance determinants in hybrid functioning model with subsurface flow operation. They are advantageous to be used as a cost-effective and environmentally friendly alternative to other WWTPs in wastewater management and it should be further explored. The aim of this study was: 1) to determine the removal efficiency (%) of total culturable, Cefixime-, Ciprofloxacin- and Linezolid-resistant bacteria, 2) to analyze bacterial inactivation by chlorination, 3) to carry out susceptibility profiling following identification of bacterial isolates by MALDI/MS, 4) to screen for Ciprofloxacin and Linezolid

antibiotics removal and 5) to conduct molecular analysis of ARGs removal by the wetlands.

Materials and methods

Designing a hybrid subsurface flow constructed wetland (HSSFCW)

A HSSFCW was constructed with three rectangular operational units with dimensions of 37×6×6 inches (length, width, and height) and a tank. All the units of the system are portrayed in Figure 1. These were connected through polyvinylchloride pipes (length = 317 mm, inner diameter = 20 mm) and arranged in a sequential manner. The media used in the first treatment unit (SS-VF) was composed of 1 inch of soil as the top layer, 2 inches of fine gravel as the middle layer and 2 inches of coarse gravel as the bottom layer. The second unit consisted of 1 inch of soil as the top layer, 2 inches of coarse gravel as the middle layer and 2 inches of pebbles settled as the bottom layer. These two units were planted with *Typha latifolia*. The last unit consisted of sand bed (SB). The right end of each pod was slightly tilted downwards to facilitate the natural gravitational pull on the water. The distance between the first and the last pod was 5 feet, and there was a 1-foot distance between each pod. Each pod had a water exit point that was 1 inch above the bottom of the pod. Before setting up working conditions, the reactor was optimized for 14 days and fed with tap water to establish plant acclimatization and stabilization of microbial growth in the plant rhizosphere. During this time, the system was operated in a batch mode with a contact time of 3 days, and no sampling was performed.

Treatment of domestic wastewater and sample collection

Prior to the beginning of every new treatment phase, an initial characterization of the influent (raw wastewater from the sewage lines of the suburban community of Quaid-i-Azam University, Islamabad, Pakistan) was performed for a period of four months (September-December) and almost every other week. A 12-liter sample of domestic wastewater was initially fed into each cycle as an influent into a tank (with a capacity of 500 L) and streamed through the wetland at a rate of 0.5 L per hour. The tank was given a holding period of nearly two hours to allow for the sedimentation of any particle materials. This partially treated wastewater was then allowed to flow through the three functional (SS-VF, SS-HF, and Sand bed) units of the mesocosm in a subsequent manner. Characterization of the effluent was done after its collection from the last processing unit using sterile 500 mL polypropylene bottles.

Bacterial Profiling of plant rhizosphere and biofilm formers in hybrid subsurface flow constructed wetland (HSSFCW)

The diverse microorganisms inhabiting the rhizosphere of *Typha latifolia* and forming biofilm on wetland media were

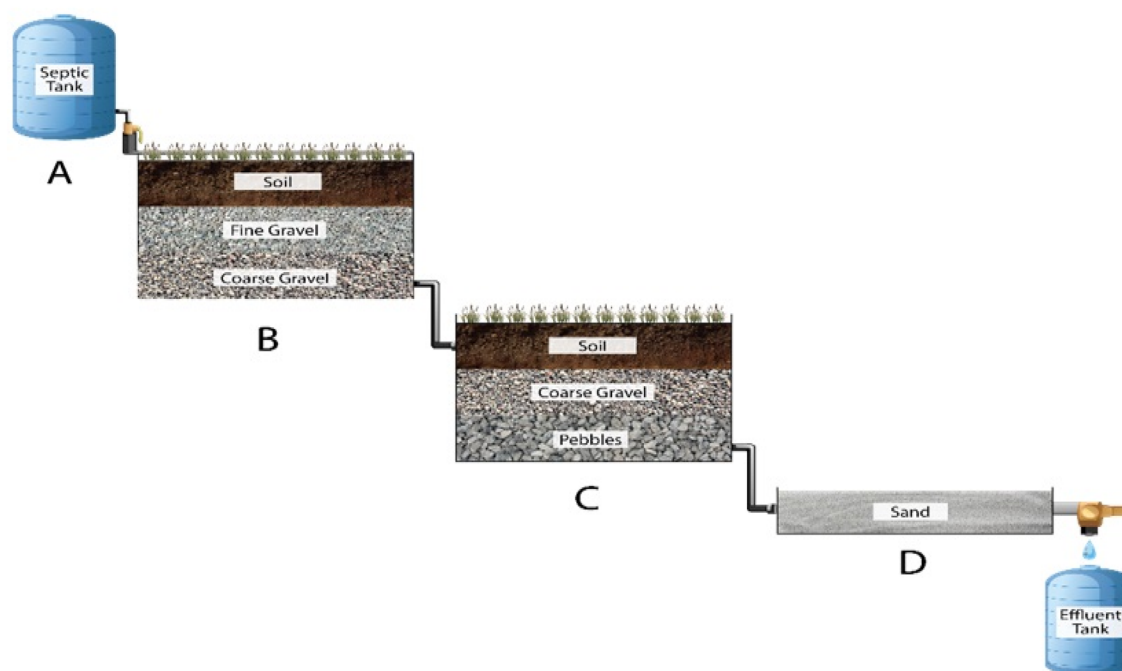


Figure 1. Schematic illustration of the mesocosm Hybrid Subsurface Flow Constructed Wetland (HSSFCW) with three operational units and two collection tanks where, A) the septic tank functions to perform the primary sedimentation of influent, B) Vertical SSF unit that primarily promotes aerobic microbial processes to treat wastewater C) Horizontal SSF unit that favors anerobic conditions of the wetland D) Sand bed for final polishing. *Typha latifolia* in Pods B and C was chosen as the plant of choice. In effluent tank, the treated wastewater was collected for recycling up till 7 days until next influent sample was run.

investigated. Soil sample (1g) collected from the roots of the plants was dissolved in sterilized water and serially diluted. From various dilutions (10^{-4} , 10^{-6} and 10^{-8}), 100 μ L was spread-plated on nutrient agar plates and bacterial counts were determined after 24h of incubation at 37°C. For bacterial profiling from biofilm, gravel stones from the wetland substrate were taken to the laboratory in a sterilized plastic bag, and the biofilm was removed. This was followed by serial dilution and spreading of the dilutions on nutrient agar plates. All isolates were later processed for identification by MALDI-TOF MS analysis.

Physicochemical analysis

Wastewater was characterized for different physico-chemical parameters, including pH, electrical conductivity (EC), total dissolved solids (TDS), and salts using their respective digital meters. 5210 B standard method (5-day test) for estimating Biochemical oxygen demand (BOD) and a quick-test kit of range 25–1500 mg L^{-1} (114,541) by Merck was used for chemical oxygen demand (COD). Quick-test kits of range 2–150 mg L^{-1} (100,683) by Merck were used for measuring Total Kjeldahl nitrogen. Total dissolved solids (TDS) in water samples were measured by the standard method 1540C. For conducting this analysis, all the reagents used were of analytical grade, and the instruments were accurate and precise.

Bacterial enumeration, isolation and inactivation by chlorination

Initial screening of resistant bacteria was carried out by serially diluting the influent and effluent samples and spreading

them on nutrient agar (Oxoid chemical company UK) plates supplemented with antibiotics (Sigma Chemicals Co, Dernstadt, Germany) ciprofloxacin (Cip) (4 $\mu\text{g/mL}$), cefixime (Cef) (1 $\mu\text{g/mL}$) and linezolid (Lzd) (4 $\mu\text{g/mL}$). For the enumeration of respective antibiotic-resistant populations, nutrient agar media was supplemented with Cef- (cephalosporin) and Cip- (quinolone) antibiotics. Blood agar supplemented with Lzd- (oxazolidinone) antibiotic were used for further isolation of *Staphylococcus* and *Streptococcus* species. Those colonies that appeared distinct were selected for further purification and characterization. To achieve an estimate of total heterotrophic bacteria, the samples were spread on nutrient agar plates without antibiotics. Further, bacterial counts were reported as $\log_{10}$ CFU/mL in all the wastewater samples. All effluent samples were subjected to chlorination using liquid sodium hypochlorite (by the DPD method) to achieve various doses (0, 0.5, 1.0, 2.0, 5.0 and 10.0 $\text{mg Cl}_2/\text{L}$) for a reaction time of 15 min. The process was terminated by adding sodium thiosulfate (1.5%). The dosage of chlorination was expressed in terms of CT values (the product of initial sodium hypochlorite concentration and contact time). Finally, disinfection of the effluent samples was carried out with CT values 0, 7.5, 15, 30, 75 and 150 $\text{mg Cl}_2 \text{ min/L}$.

MALDI-TOF/MS analysis of isolated strains

A total of 64 antibiotic-resistant bacteria and 5 isolates obtained from plant rhizosphere and wetland biofilm were identified to determine their species by matrix-assisted laser desorption/ionization time-of-flight mass spectrometry MALDI-TOF/MS. The MALDI-TOF spectrum against each bacterial isolate was generated in the mass spectrometer

(Bruker, Germany) and imported directly into the MALDI Biotyper® software. The analysis was carried out with default settings by matching the spectra of each isolate against the spectra of reference species present within the database library. Species identification was then presented with a score of reliability. Standard Bruker criteria were applied for interpreting the score; acceptable scores for specie-level identification were ≥ 2.0 , while that for genus identification were ≥ 1.7 but ≤ 2.0 .

Antimicrobial susceptibility testing

Antibiotic-resistant bacteria resistant to Cip, Cef and Lzd isolated from the effluent samples were subjected to antimicrobial susceptibility testing (AST) using the standard Kirby-Bauer disk diffusion method. Inoculum suspension was prepared by mixing 24 h fresh bacterial culture in 5 mL normal saline whose turbidity was adjusted to 0.5 McFarland standard. The suspension was spread on Mueller-Hinton agar using sterile cotton swabs to create a bacterial lawn. The antibiotics and their respective disk potencies used against gram negative strains included gentamicin (10 µg), aztreonam (30 µg), piperacillin/tazobactam (100/10 µg), imipenem (10 µg), ceftriaxone (30 µg) and minocycline (30 µg). Meanwhile, chloramphenicol (30 µg), azithromycin (30 µg) and ciprofloxacin (5 µg) were used against gram positive strains. Antibiotic susceptibility testing was done by measuring the zone of inhibition after 24 h of incubation at 37°C. Classification was done into three groups, sensitive, intermediate and resistance. The strain of *E. coli* ATCC 25922 was referenced as the control during this test. Multidrug Resistance (MDR) was considered when resistance against three or more antimicrobial drugs was recorded.

Quantification of antibiotics

In this study, Cip and Lzd were screened for removal through HSSFCW in the influent and effluent samples. SCION 6000 HPLC system equipped with a UV detector 6410 was used. C18 column (4.6 mm × 150 mm) was used for running samples, and data were collected using Compass CDS software. For Cip, the mobile phase consisted of 0.025 M phosphoric acid adjusted with triethylamine at pH 3 and acetonitrile (13:87). For Lzd, the mobile phase consisted of water with 0.1% formic acid: acetonitrile 70:30 (v/v). The column temperature and pressure were maintained at 35°C and 10.57 MPa, respectively. The sample was injected at a volume of 10 µL, and the flow rate of the mobile phase was 1.0 mL/min for the detection of each antibiotic. The detection of Cip and Lzd was monitored at a wavelength of 278 and 254 nm, respectively. While the calibration graphs were plotted by preparing standard solutions of each antibiotic in the concentrations 10, 25, 50, 100 and 1000 mg/L.

Quantification of ARGs

The genetic material of the microbial contaminants, DNA was extracted from wastewater samples and analyzed by gel

electrophoresis and Nano-Drop 2000 Spectrophotometer for both quality and quantity, respectively. Real-time quantitative PCR (qPCR) methods using SYBR Green Real Time QPCR Kit (Thermo scientific, US) were used to determine the abundance of the ARGs. Temperature conditions for the quantification included initial denaturation at 95°C for 10 min, followed by 40 cycles of 30 s at 95°C and 1 min at 60°C. Subsequently, the plasmid DNA was diluted in 10-fold series to generate calibration standard curves. All the samples were run in duplicates. The quantification of ARGs was expressed as absolute abundance (copies/mL) and relative abundance (target gene copies/16S rDNA gene copies). The primer sequences are listed in Table S1.

Statistical analysis

The removal efficiency (%) of the HCW for all the measured parameters was calculated using Equation (1):

$$\% \text{Removal efficiency} = (C_{\text{influent}} - C_{\text{effluent}}) / C_{\text{influent}} * 100$$

where C_{influent} is the concentration in influent (mg/L) and C_{effluent} is the concentration in effluent (mg/L).

Wilcoxon signed-rank test was applied for evaluating the significant difference in % removal of physicochemical, microbial and antibiotic contaminants using the SPSS (Ver 19) software. The removal efficiencies (%) of the ARGs were compared by One-way ANOVA, followed by Tukey's post hoc test. Significance was considered when the *p*-value was < 0.05 .

Results and discussion

Physicochemical analysis of wastewater before and after the treatment

Hybrid subsurface flow constructed wetland (HSSFCW) has improved the physicochemical parameters of wastewater as described in the Table 1. However, no profound effect on the pH values among untreated and treated samples was observed during the operational phase of the HSSFCW and it has remained within the range of 7.03–8.32 with % fluctuation of 0.08–12.51%. High pH of soil, specifically when it exceeds 8.5, impacts soil alkalinity and negatively affects plant growth by inhibiting root development and nutrient absorption (Yang *et al.* 2024). All other parameters were improved in the effluents when compared to the influents.

Table 1. The mean concentrations of physicochemical parameters of influents and effluents in domestic wastewater.

Wastewater parameters	Influent ± SD	Effluent ± SD
pH	7.588 ± 0.03	7.465 ± 0.02
TDS	581.72 ± 1.76 mg/L	314.04 ± 1.79 mg/L
EC	640.31 ± 1.66 µS/cm	354.85 ± 1.86 µS/cm
Salts	221.66 ± 1.22 mg/L	151.66 ± 1.40 mg/L
TKN	23.58 ± 1.05 mg/L	15.27 ± 0.76 mg/L
COD	825.14 ± 2.31 mg/L	182.25 ± 0.96 mg/L
BOD	363.99 ± 1.16 mg/L	80.66 ± 0.47 mg/L

A removal efficiency of 62.89, 57.4, 36.71% for TDS, EC and salts was achieved, respectively. Examination of all parameters were conducted in triplicates and temperature was constantly monitored during the running of the reactor. Nutrient removal was examined through reduction in Total Kjeldahl Nitrogen (TKN) and the mean TKN was 23.37 and 15.27 mg/mL in influent and effluent samples, respectively. A highly significant removal ($p < 0.001$) attained through HSSFCW can be due to increased residence time in the wetland supporting greater rates of nitrification and denitrification processes.

Dissolved organic matter is recognized as a serious problem of water quality (Shafiquzzaman *et al.* 2020). The COD and BOD₅ for untreated samples were observed to have a high mean value of 825.14 and 363.99 mg/L in comparison with the U.S. EPA standards (5–8 and 8–10 mg/L) respectively. These high influent values suggested the greater degree of organic contamination of domestic wastewater. The treated samples had an average COD and BOD of 182.25 and 80.66 mg/L with wetland achieving a removal efficiency of 89.00 and 88.90%, respectively. Significant reductions in COD and BOD levels ($p < 0.006$ and $p < 0.008$, respectively) in the wastewater were observed after treatment. This significant removal can be attributed to the synergistic effects of biological and physical processes such as filtration and microbial degradation on the organic matter of wastewater during wetland operation (Sudarsan *et al.* 2018). The results obtained in this study reconfirmed the significant removal of physical and chemical contaminants from a previously published study by Sehar *et al.* (2013), employing a similar design of HSSFCW, which achieved a removal efficiency of COD by nearly 100% and BOD by approximately 99% at HRT of 8 days, i.e., 1 day longer than our study's 7 days retention time (Sehar *et al.* 2013).

Further, the percent removal efficiency with time showed an inconsistent trend for physical contaminants while indicated an increasing trend for nutrient and organic removal by HSSFCW. This improvement might be primarily due to the eventual biofilm formation and the ability of rhizosphere microorganisms present along the dense root structure of *Typha latifolia* to metabolize, absorb or degrade contaminants through phytoremediation (Martínez 2023).

Bacterial Profiling of plant rhizosphere and wetland biofilm

Typha latifolia plants were readily available in nearby sewage areas and the dead plants could be easily replaced with newer ones to maintain the density of the plantation in the reactor. A dense root hair structure within the wetland substrate was detected and from which three distinct bacterial isolates were isolated and purified. They were identified as *Enterobacter cloacae*, *Serratia liquefaciens* and *Serratia odorifera*. Notably, Kelty *et al.* (2012) and Vuotto *et al.* (2014) investigations also reported that *Serratia* and *Enterobacter* species observed in the plant rhizosphere have the ability to sequester pollutants, thereby improving the wetland efficiency. While biofilm-forming bacterial isolates on wetland

media were identified as *Acinetobacter baumannii* and *Staphylococcus spp.* The presence of *Acinetobacter* and *Staphylococcus* species were also previously confirmed in the wetland system by Hou *et al.* 2017, thus further supporting our findings. The identification of these bacterial species highlights their role in benefitting plant growth, promoting nutrient uptake, and facilitating contaminant breakdown.

Removal of total heterotrophic and antibiotic-resistant bacteria

The mean concentration of antibiotic-resistant bacteria in the influent samples (raw domestic wastewater) was 5.75 ± 0.14 , 5.96 ± 0.21 and 4.45 ± 0.55 log₁₀ CFU/mL for Cef, Cip and Lzd respectively. While, the mean concentration of the total heterotrophic bacteria (THB) was 7.72 ± 0.16 log₁₀ CFU/mL in untreated wastewater, HSSFCW was able to reduce it to an average concentration of 5.24 ± 0.59 log₁₀ CFU/mL in the treated wastewater. A mean difference of 5.56×10^7 (2.48 log) CFU/mL was observed for THB. The mean concentration counts of Cef-resistant bacteria reduced to 3.99 ± 0.34 log₁₀ CFU/mL in the effluents with an average difference of 5.75×10^5 (1.75 log) CFU/mL during the wastewater treatment. Similarly, mean effluent concentration of Cip-resistant bacteria was reduced to 4.17 ± 0.22 log₁₀ CFU/mL and Lzd-resistant bacteria to 2.74 ± 0.17 log₁₀ CFU/mL. Meanwhile, Figure 2 indicates the reduction in log₁₀ CFU/mL concentration of THB, Cef-, Cip-, and Lzd-resistant bacteria in the domestic wastewater after treatment by the HSSFCW during a 16-weeks running time.

Pathogen removal from wastewater employing horizontal sub-surface flow CW by Morató *et al.* (2014) has reported a removal ratio of 1.4–2.2 and 1.2–2.2 log units for fecal *Enterococci* and total coliforms, respectively. In comparison, our system of HSSFCW has achieved a similar (for ARBs) and even higher (for THB) removal ratio primarily due to the efficient microbial degradation in the aerobic, anoxic, and anaerobic conditions created in the reactor and enhanced oxygen transfer due to the additional vertical flow combined with horizontal flow of wastewater in the system. The log₁₀ removal ratios during the 16-week period for wastewater treatment by HSSFCW is presented in Figure 3. This improved performance of hybrid systems over traditionally designed wetlands is also highlighted by the removal ratio of pathogens reported by Kadlec and Wallace (2008). These ratios ranged from 1 to 3 log units. Even though different configurations of constructed wetlands have a certain potential to remove pathogens in wastewater, these removal values for pathogenic bacteria are typically in the range 0.01 to 5.68 log for HSSFCW, 0.07 to 5.3 log₁₀ for free-water surface (FWS) and 0.35 to 6.08 log for vertical flow (VF) CWs (Paraskevopoulos and Smeets 2019). The current study attained a 1.72 to 1.78 log₁₀ removal for cip- and lzd-resistant bacteria which also consolidates the findings of 1.5 to 2.5 log₁₀ removal ratios reported by Ma *et al.* (2020) for ARBs (fluoroquinolone- and linezolid-resistant bacteria respectively). Luo *et al.* (2010) found a greater 2–3 log reductions in ARBs in their hybrid CW system, which was slightly

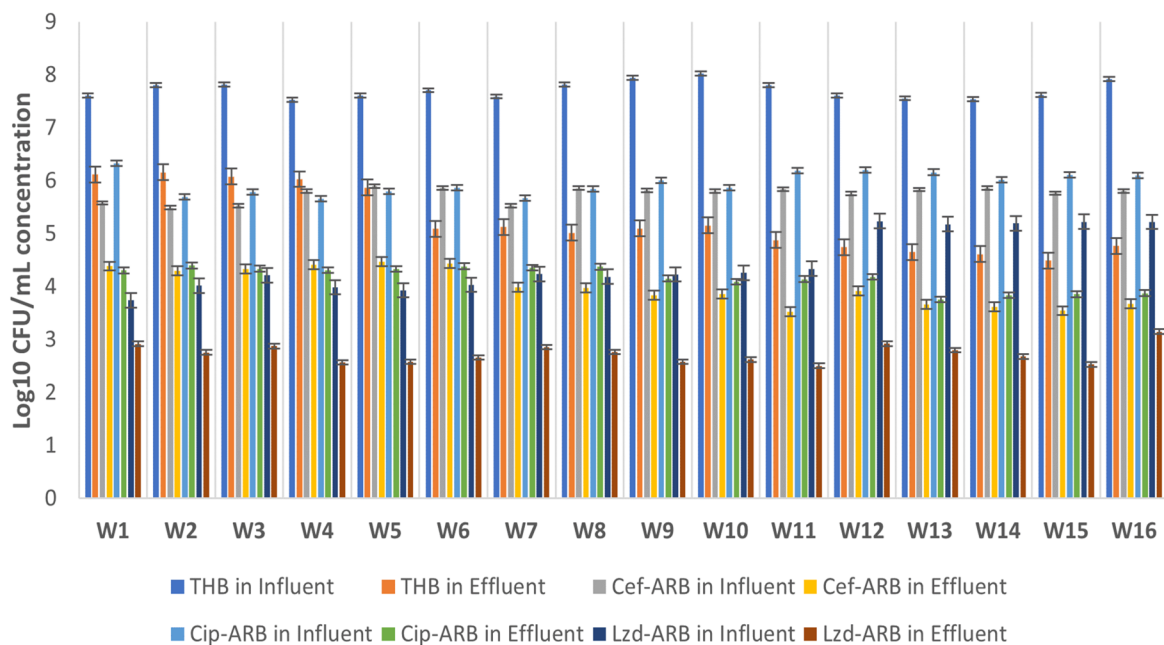


Figure 2. Removal of total heterotrophic bacteria and antibiotic-resistant bacteria (cefexime-resistant, ciprofloxacin-resistant, and linezolid-resistant) before and after treatment across 16 weeks. This bar chart represents the log₁₀ concentration units (CFU/mL) of bacterial population including total heterotrophic bacteria (THB) and antibiotic-resistant bacteria (ARB) resistant to cefexime (Cef-ARB), ciprofloxacin (Cip-ARB), and linezolid (Lzd-ARB) in wastewater samples over the period of 16 weeks (W1–W16).

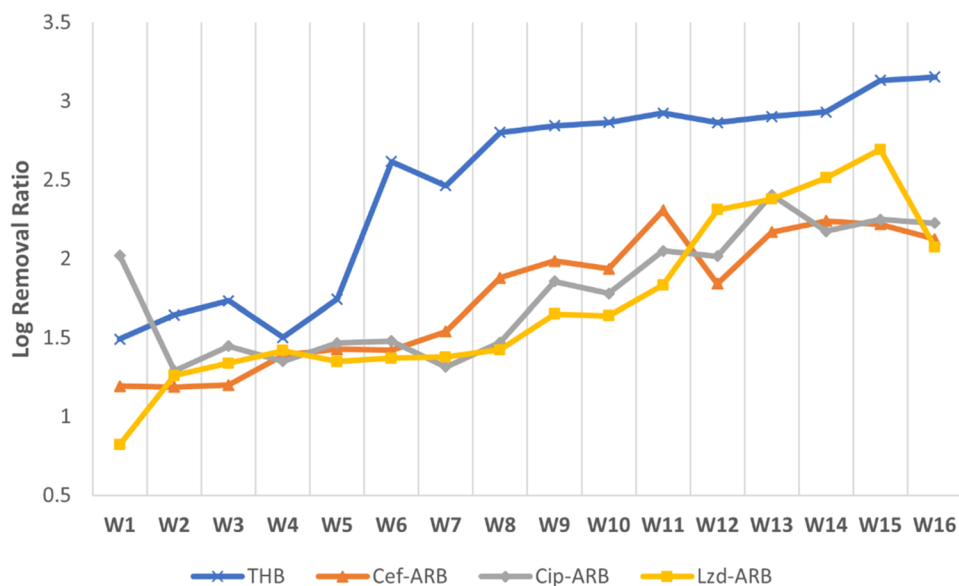


Figure 3. Log removal ratio of total culturable bacteria and antibiotic-resistant bacteria during the treatment Week1 (W1) to week 13 (W13).

higher than compared to our results but is still in the expected range for resistant bacterial species, which otherwise are more difficult to remove. The lower removal ratios for ARBs compared to THB might be due to their increased resistance mechanisms, allowing them to withstand the wastewater treatment through natural and engineered processes that occur in wetlands (Rizzo *et al.* 2013).

Furthermore, the removal efficiency (%) of the HSSFCW for THB was within the range of 96.78–99.93%. Similarly, the wetland removed ARBs with an efficiency of 93.52–99.42% for Cef-ARB, 94.87–99.61% for Cip-ARB and 84.91–99.79% for Lzd-ARB. A highly significant removal

($p < 0.001$) was observed for THB, Cef-, Cip- and Lzd-resistant bacteria. The enhanced removal efficiency of the reactor can be attributed to the increased retention time period of wastewater in the wetland technology compared to other conventional WWTPs. Horizontal SSFCWs operating at HRT of 1.5, 3, 5 and 6 days and planted with *Typha angustifolia* have shown a removal efficiency (%) of 0.8, 1.7, 2.3 and 2.4 log₁₀ CFU/100 mL respectively for total coliforms (Khatiwada and Polprasert 1999). This also validates the water retention period of 7 days in our study as an optimized time period for removal of microbial contamination in wastewater.

Chlorine inactivation of resistant bacteria in treated wastewater

Although chlorination has been widely used as a disinfection method, its efficacy in the face of rising antibiotic resistance needs to be critically assessed. The results of chlorination disinfection presented in Figure 4 suggested that all ARBs were effectively inactivated with increased dosage of chlorine. No colony was observed for any resistant bacteria above 30 mg Cl₂ min/L of chlorine dosage. The same was observed at 75 and 150 mg Cl₂ min/L. Cef-resistant bacteria showed tolerance to chlorination at 7.5 mg Cl₂ min/L in 5 and at 15 mg Cl₂ min/L in 2 effluent samples, respectively. While Cip-resistant bacteria showed tolerance at 7.5 mg Cl₂ min/L in 3 effluent samples. Lzd-resistant bacteria did not exhibit tolerance to chlorination at any dosage. Our research findings align well with and outperformed those of Yuan *et al.* 2015, where resistant bacterial population was effectively inactivated, except for Sulfadiazine (SD-) and Erythromycin (ERY-) resistant bacteria. These were not completely eliminated until the dosage reached 60 mg Cl₂ min/L, while in our studies the maximum dosage to which resistant bacterial population have displayed tolerance was at 30 mg Cl₂ min/L.

Microbial identification and susceptibility profiling

A total of 64 bacterial isolates from the treated wastewater samples were obtained by repeated streaking to obtain pure colonies. Out of these 64 antibiotic-resistant bacteria as described in Table 2, 24 (37.5%) were resistant to Cip, 24 (37.5%) to Cef and 16 (25%) to Lzd. The most frequently detected ARBs in the domestic wastewater against cefixime were found to be *E. coli* 9 (37%) followed by *Stenotrophomonas maltophilia* 5 (21%), *Acinetobacter baumannii* 3 (13%), *Enterobacter cloacae* 3 (13%), *Serratia odorifera* 2 (8%), *Enterobacter aerogenes* 1 (4%) and *Chryseobacterium*

meningosepticum 1 (4%). Similarly, against ciprofloxacin, 10 (42%) *E. coli*, 5 (20%) *Stenotrophomonas maltophilia*, 3 (13%) *Klebsiella pneumoniae*, 3 (13%) *Staphylococcus aureus*, 2 (8%) *Enterobacter cloacae* and 1 (4%) *Serratia marcescens* species were detected. 11 ARBs found against linezolid included 11 (69%) *Staphylococcus aureus*, 3 (19%) *Streptococcus pyogenes* and 2 (12%) *Streptococcus pneumoniae* species.

A total of 45 Gram-negative bacterial species belonging to the genera *Escherichia*, *Stenotrophomonas*, *Klebsiella*, *Acinetobacter*, *Serratia* and *Enterobacter* isolated were tested against six antibiotics and a total of 19 Gram-positive bacterial species of the genera *Staphylococcus* and *Streptococcus* were tested for their antimicrobial sensitivity pattern against three antibiotics. Figure 4 exhibits the antibiotic susceptibility pattern of the (a) Gram-negative and (b) Gram-positive bacterial isolates obtained in this study.

Resistance to ceftriaxone was observed in 31 isolates (68.88%), Piperacillin/Tazobactam 18 isolates (40%), Imipenem 31 isolates (68.88%), Gentamicin 32 isolates (71.11%) and Minocycline 8 isolates (17.77%) and all of

Table 2. Antibiotic-resistant bacteria isolated from effluents of each reactor cycle.

ARB	Resistance against Ciprofloxacin	Resistance against Cefixime	Resistance against Linezolid
<i>E. coli</i>	10 (42%)	9 (37%)	
<i>Stenotrophomonas maltophilia</i>	5 (20%)	5 (21%)	
<i>Acinetobacter baumannii</i>	–	3 (13%)	
<i>Enterobacter cloacae</i>	2 (8%)	3 (13%)	
<i>Enterobacter aerogenes</i>	–	1 (4%)	
<i>Serratia odorifera</i>	–	2 (8%)	
<i>Serratia marcescens</i>	1 (4%)	–	
<i>Klebsiella pneumoniae</i>	3 (13%)	–	
<i>Chryseobacterium meningosepticum</i>	–	1 (4%)	
<i>Staphylococcus aureus</i>	3 (13%)	–	11 (69%)
<i>Streptococcus pyogenes</i>	–	–	3 (19%)
<i>Streptococcus pneumoniae</i>	–	–	2 (12%)

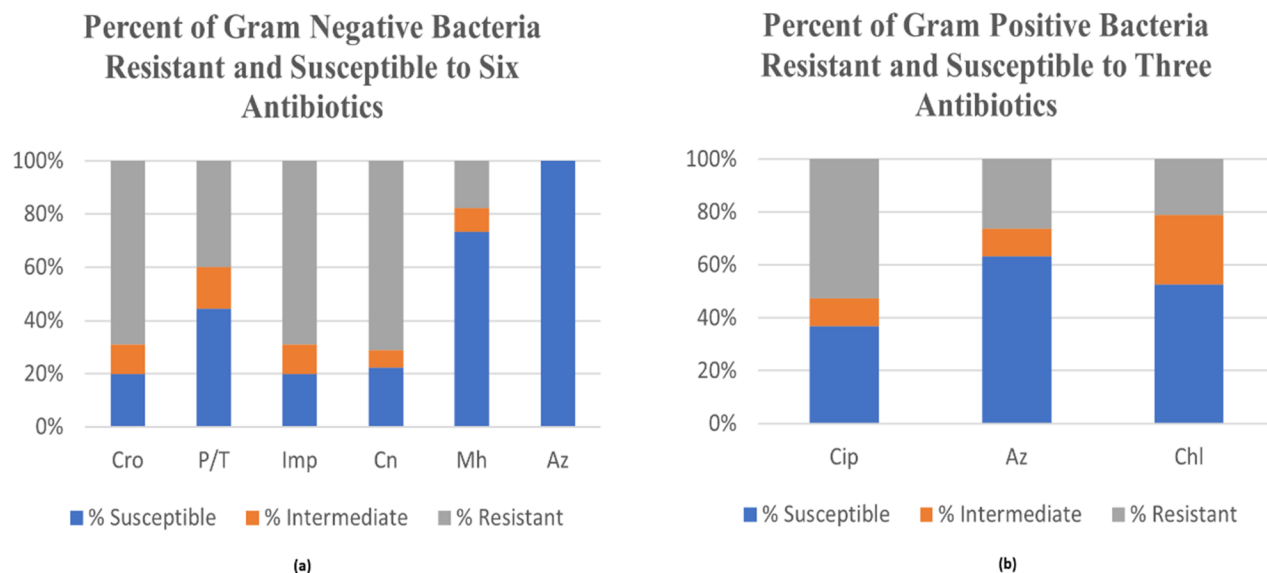


Figure 4. Results of the antibiotic susceptibility test for (a) Gram-negative ($n=45$) and (b) Gram-positive ($n=19$) bacterial isolates where, Mh: Minocycline, Cro: Ceftriaxone, Az: Aztreonam, Cn: Gentamicin, P/T: Piperacillin/tazobactam, Imp: Imipenem, Chl: Chloramphenicol, Cip: Ciprofloxacin and Az=Azithromycin.

them were Gram-negative bacterial isolates. Isolates susceptible to Ceftriaxone, Piperacillin/Tazobactam, Imipenem, Gentamicin, Minocycline were found to be as: 9 (20%), 20 (44.44%), 9 (20%), 10 (22.22%) and 33 (73.33%), respectively. All 45 Gram-negative isolates (100%) were susceptible to Aztreonam. Among the 19 Gram-positive isolates, 10 isolates (52.63%) tested resistant against Ciprofloxacin, 5 isolates (26.31%) against Azithromycin and 4 isolates (21.05%) against Chloramphenicol.

These results also suggested that among the total 64 bacterial isolates, 12 (18.75%) isolates were sensitive and resistant to at least one antibiotic tested, and 4 isolates (6.25%) were resistant to all antibiotics tested with 28 isolates (43.75%) resistant to 3 or 4 antibiotics and were categorized as multidrug-resistant (MDR) strains.

Antibiotic removal by the hybrid subsurface flow constructed wetland (HSSFCW)

The performance of HSSFCW in removing two different antibiotics, such as ciprofloxacin and linezolid was evaluated. Fluoroquinolones (including ciprofloxacin) and oxazolidinones (such as linezolid) are frequently detected in water bodies, and conventional WWTPs do not efficiently remove these compounds (Kümmerer 2009). Figure 5 demonstrates the concentration of antibiotics in a total of twenty-four samples (12 untreated and 12 treated) analyzed by HPLC and the reactor's efficiency in removing them from the domestic wastewater. The highest concentration of Cip and Lzd detected in raw wastewater samples was 2400 and 92.4 ng/L. Further, Cip was detected in every sample of influent and effluent while Lzd was detected in all influent samples but remained undetected in three effluent samples, which suggested its complete removal. Mean concentrations

of Cip and Lzd in influent samples were 1044.25 and 61.75 ng/L, respectively, and reduced in effluent samples to 369.83 and 6.19 ng/L, respectively.

The concentration of Lzd in urban wastewater was reported to be in the concentration range of 4–171 ng/L, 1–50 ng/L in the surface water of Spain and below 11.5 ng/L in Changxing Island, China (Jiang *et al.* 2018). These research findings report a mean concentration of 61.75 ng/L of Lzd in domestic wastewater indicating the notable presence of oxazolidinone antibiotics that are not completely degraded. This finding underscores the abundance of residual antibiotics in domestic wastewater contributing to aquatic and environmental contamination. A study investigating the presence of antibiotics in treated wastewater sampled from multiple European WWTPs also found the highest concentration of antibiotics belonging to the class of fluoroquinolones. The highest concentration was observed in the wastewater of Portugal and was reported up to 1435.5 ng/L of unmetabolized ciprofloxacin (Rodriguez-Mozaz *et al.* 2020). The current investigation of 12-week samples, established a similar but slightly higher range of approximately 370–1420 ng/L for ciprofloxacin (fluoroquinolone antibiotic).

In a comparative analysis of antibiotic abundance in wastewater samples from different locations of Rawalpindi and Islamabad in Pakistan, the highest concentration detected was of Ciprofloxacin compared to other antibiotics. Its concentration ranged from 332.155 µg/mL at site Nullah Lai in Rawalpindi to 0.157 µg/mL at Nullah Lai site in Islamabad (Zafar *et al.* 2021). Higher concentration of Cip (1044.25 ng/L) detected in the domestic wastewater from the sampling site in the city of Islamabad for the current investigation also underlines its common consumption patterns and improper disposal practices of antibiotics in the area reflecting the dire need for efficient wastewater management.

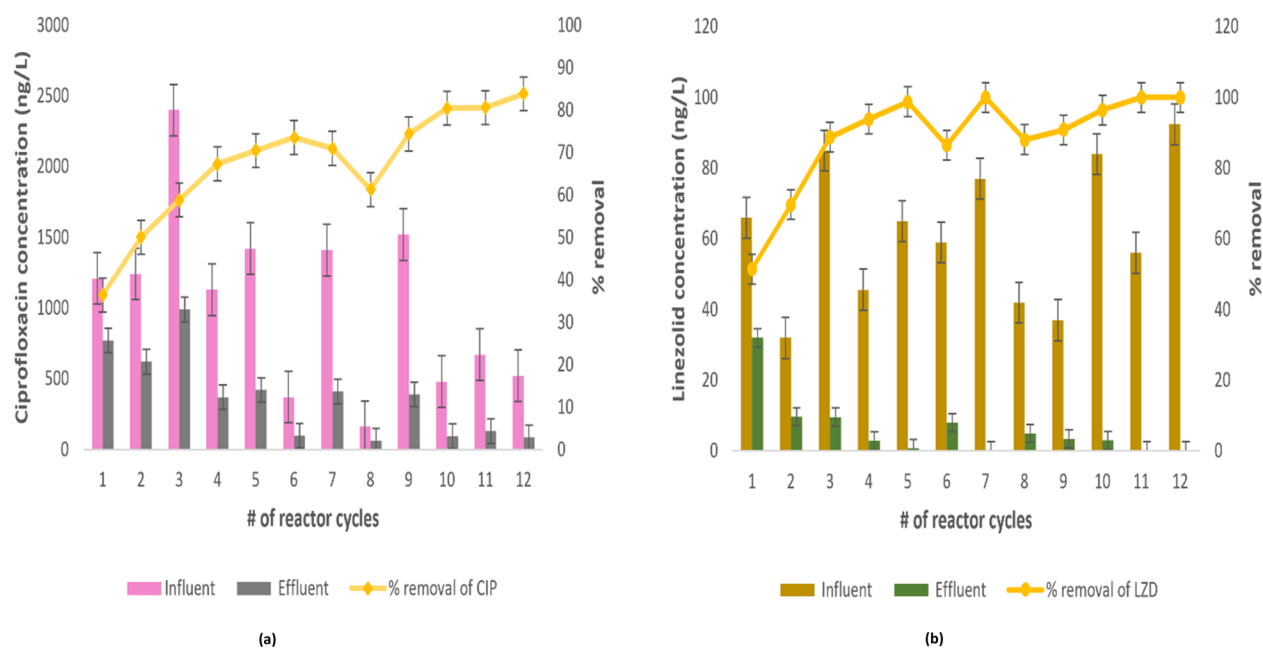


Figure 5. Concentration of antibiotics a) Ciprofloxacin and b) Linezolid in domestic wastewater pre- and post-treatment during 12-week cycles of wetland operation.

The wetland efficiency in removing the antibiotics improved with time during the 12-week course of running. It was observed to be the highest during 12th week (83.85%) for Cip as compared to 36.36% during week 1. 100% removal efficiency was attained for Lzd by the Hybrid CW during 7th, 11th and 12th week as compared to 51.52% during week 1 of the reactor cycle. A decrease in average concentration of Cip was observed to be 674.42 ng/L and 55.59 ng/L for Lzd pre- and post-treatment. Lzd is less frequently studied in wastewater, which makes our findings a valuable contribution in the limited data available on its removal in the environment. Although de Andrade *et al.* (2021) indicated that removal efficiencies varied according to certain wetland parameters and types of antibiotics, our research shows that the HSSFCW outperforms other systems in attaining higher removal efficiency (%) for antibiotics, especially for linezolid.

Constructed wetlands perform better in removing contaminants as they mature, with the development of more robust microbial biofilms, denser growth and increased plant uptake and soil and sediment accumulation improves with time (Zhang *et al.* 2014). Matamoros *et al.* (2007) also reported improved removal efficiencies for antibiotics and other pollutants in wetlands running for extended time periods. A significant difference in antibiotic concentration between untreated and treated wastewater was observed for Cip ($p < 0.001$) and Lzd ($p = 0.002$). Bacteria in the wetland's biofilm degrade these antibiotic residues through their metabolic pathways to utilize them as a carbon source. Another potential reason for observing this significant reduction of antibiotics concentrations before and after treatment can be attributed to their enhanced photodegradation in the open water spaces of CWs (Kim 2010).

Reduction of antibiotic-resistant genes by the hybrid subsurface flow constructed wetland (HSSFCW)

Resistance to beta-lactam antibiotics is particularly increasing and the widespread occurrence of these resistant genes

needs to be documented in environmental settings. This study focused on the abundance of three β -lactamase genes *bla*CTX, *bla*TEM and *bla*OXA and their removal from domestic wastewater using constructed wetlands, providing insights into the wetland's potential to eliminate ARGs and mitigate their spread. Figure 6 shows the mean relative abundance of ARGs normalized to 16S *rDNA* in the untreated and treated wastewater.

The domestic wastewater contained an average relative abundance of 3.83×10^9 copies/mL of 16S *rDNA* indicating a high level of total heterotrophic bacterial contamination. This is in accordance to the range (10^8 and 10^9 copies/mL) of 16S *rDNA* abundance reported for urban wastewater (Zhang *et al.* 2020). The influent abundance of ARGs was the highest for *bla*TEM (2.9×10^6 copies/mL), followed by *bla*CTX (5.54×10^4 copies/mL) and the lowest for *bla*OXA (5.26×10^3 copies/mL). This difference in abundance of β -lactamase genes observed in the present study suggesting lower concentration levels of *bla*CTX genes compared to *bla*TEM is in accordance with Rodriguez-Mozaz *et al.* (2015). This is probably due to the greater selective pressure exerted by extended-spectrum beta-lactam (ESBL) antibiotics that are targeted by *bla*TEM. The treated wastewater had a mean relative abundance decreased to 1.18×10^9 copies/mL for 16S *rDNA*, 4.37×10^5 copies/mL for, and 1.35×10^4 copies/mL for *bla*CTX and to 1.73×10^3 copies/mL *bla*OXA.

Figure 7 shows the plot of CT values for 16S *rDNA* and ARGs in both untreated and treated wastewater samples. The genes analyzed include β -lactamase genes CTX, TEM, OXA (the CT mean of OXA gene after cycle 8 was above the limit of detection and was excluded in the plot), and 16S *rDNA*. Boxes represent the interquartile range (IQR) of the CT values, with the upper whisker representing the 75th percentile and lower whisker representing the 35th percentile. The median value is indicated by the horizontal line inside each box. The mean CT value of 5.4 in influent and 7.6 in effluent was recorded for 16S *rDNA*. The increase in the mean CT value for TEM after wastewater treatment was obtained at 21.6 compared to 17.8 in the influents. Similarly,

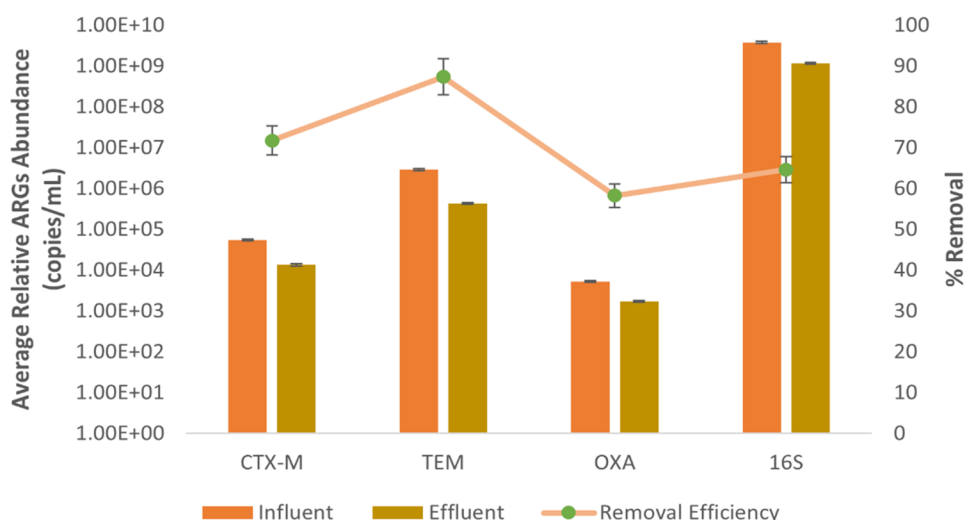


Figure 6. Reduction of ARGs and 16S *rDNA* gene in wastewater after treatment through hybrid subsurface flow constructed wetland (HSSFCW). The mean relative concentration of β -lactamase genes CTX, TEM and OXA (ARGs copies/16S *rDNA* copies) and 16S *rDNA* are plotted in the influents and effluents. Percentage removal (%) of these genes by the HSSFCW is depicted by the line graph.

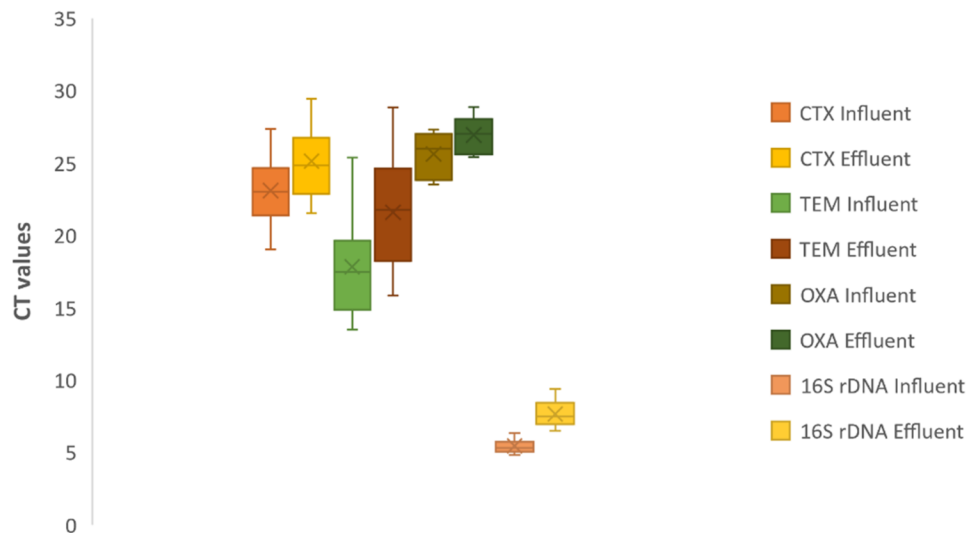


Figure 7. Box and whisker plot of CT values of ARGs and 16S rDNA in influent and effluent wastewater samples.

the mean CT value increased to 25.1 from 23.1 for CTX and for OXA, it increased from 25.6 to 26.9 post-treatment. The HSSFCW demonstrated a significant removal efficiency (%) for ARGs in the wastewater samples, with a p value of 0.001. Comparative analysis of the removal efficiency (%) of the reactor for these genes showed no significant differences between CTX and OXA ($p=0.161$), CTX and 16S rDNA ($p=0.886$), TEM and 16S rDNA ($p=0.197$). Removal efficiency (%) significantly differed between CTX and TEM ($p=0.042$), TEM and OXA ($p<0.001$) and OXA and 16S rDNA ($p=0.040$).

Constructed wetlands perform better in removing ARGs from wastewater compared to conventional WWTPs which on the contrary, achieve partial or no removal of ARGs. Different configurations of CWs exhibit varying removal efficiency (%) for these genes in wastewater depending upon the wetland design and operating conditions. As opposed to subsurface flow wetlands, lower removal efficiency of 50–70% for ARGs are reported for surface flow beds. This removal is primarily attributed to sedimentation rather than microbial degradation. Subsurface flow having horizontal flow cells can remove resistant genes up to 84% and those employing vertical flow cells can remove them up to 90% (Chen *et al.* 2022). A removal efficiency of 89.6% for CTX, 98.4% for TEM, and 73.4% for OXA was observed in our study. This wetland configuration, which combines the synergistic effect of horizontal and vertical flow of wastewater allowed for the cumulative degradation of genetic material under both anaerobic and aerobic conditions. Wang *et al.* (2020) achieved a removal efficiency of 90% for ARGs such as *bla*TEM and *sul*2 with Hybrid CW, with better performance of wetlands with greater diversification of plants and higher retention time. The optimized HRT was >5 days and plant species included *Phragmites australis* and *Typha latifolia* demonstrating consistency with this study's removal efficiency for *bla*TEM but higher than what we observed for CTX and OXA. Similar rates of gene removal observed in this study could be due to the same choice of plants (*Typha latifolia*) and HRT of 7 days.

Conclusion

In conclusion, the hybrid subsurface flow constructed wetland (HSSFCW) was efficient in improving the physicochemical properties of wastewater, reducing the microbial load, lowering the concentration abundance of antibiotics and antibiotic-resistant genes. A significant proportion of the bacterial isolates found in the raw domestic wastewater were multidrug-resistant, and the majority of the isolates exhibited resistance to routinely prescribed antibiotics. The presence of these pathogens in the rhizosphere and biofilm suggests their beneficial role in enhancing wastewater treatment processes. Recognized for their role in promoting nutrient cycling, breaking down organic matter, and supporting contaminant removal, these microbial communities can be further examined for their potential mechanism in degrading antimicrobial agents and genetic material (ARGs). Ciprofloxacin was present in greater concentrations compared to linezolid while the % removal for both the antibiotics through HSF CW was nearly the same. The results of these studies are in accordance with the previously published research articles on the prevalence of emerging contaminants, highlighting their widespread occurrence and a source of contamination in the wastewater of urban areas which are not eliminated through conventional WWTPs. The HSSFCW can be upgraded into larger-scale wastewater treatment facilities in an effort to mitigate the spread of antibiotic resistance. This will decrease the likelihood of persistent contaminants, such as ARB and antibiotics, in the treated water and reduce their transmission to the public, farms and the general water bodies. Treated wastewater can be an alternative water source that could be used for irrigation to ensure safe water for health and environment by following further risk assessment protocols and considering potential detrimental effects. These findings can be used as a guideline for wastewater treatment and ensure safe practices for both public health and agricultural use.

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Disclosure statement

There are no conflicting interests to disclose, according to the authors.

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References

- Biswal BK, Balasubramanian R. 2022. Constructed wetlands for reclamation and reuse of wastewater and urban stormwater: a review. *Front Environ Sci*. 10. doi: [10.3389/fenvs.2022.836289](https://doi.org/10.3389/fenvs.2022.836289).
- Chen P, Yu X, Zhang J, Wang Y. 2022. New and traditional methods for antibiotic resistance genes removal: constructed wetland technology and photocatalysis technology. *Front Microbiol*. 13:1110793. doi: [10.3389/fmicb.2022.1110793](https://doi.org/10.3389/fmicb.2022.1110793).
- Chen Y, Li Y, Yang S, Chiang TY, Zhu X, Hu J. 2022. Controlling biofilm growth and its antibiotic resistance in drinking water by combined UV and chlorination processes. *Water*. 14(22):3643. doi: [10.3390/w14223643](https://doi.org/10.3390/w14223643).
- da Silva AMR, dos Santos DR, de Andrade LFE. 2022. Constructed wetlands for the removal of antibiotic-resistant bacteria and genes from hospital wastewater. *J Phytoremediat*. 24(2):102–115. doi: [10.1080/1547691X.2022.2036781](https://doi.org/10.1080/1547691X.2022.2036781).
- de Andrade LFE, dos Santos DR, da Silva GF, de Lima AD. 2021. Effectiveness of constructed wetlands for the removal of antibiotics from wastewater. *J Phytoremediat*. 23(3):245–258. doi: [10.1080/1547691X.2021.1897115](https://doi.org/10.1080/1547691X.2021.1897115).
- De Lille MV, Cardona MAH, Xicum YT, Giacomán-Vallejos G, Quintal-Franco C. 2021. Hybrid constructed wetlands system for domestic wastewater treatment under tropical climate: effect of recirculation strategies on nitrogen removal. *Ecol Eng*. 166:106243. doi: [10.1016/j.ecoleng.2021.106243](https://doi.org/10.1016/j.ecoleng.2021.106243).
- Dires S, Birhanu T, Ambelu A, Sahilu G. 2018. Antibiotic resistant bacteria removal of subsurface flow constructed wetlands from hospital wastewater. *J Environ Chem Eng*. 6(4):4265–4272. doi: [10.1016/j.jece.2018.06.034](https://doi.org/10.1016/j.jece.2018.06.034).
- Haider H, Ghumman AR, Al-Salamah IS, Alharbi A, El-Ghoul Y, Martuza MA. 2024. A strategic approach to sustainable wastewater management and reuse in Saudi Arabia. *Environ Res Commun*. 6(7):075009. doi: [10.1088/2515-7620/ad58af](https://doi.org/10.1088/2515-7620/ad58af).
- He Y, Zhang L, Jiang L, Wagner TV, Sutton NB, Ji R, Langenhoff A. 2021. Improving removal of antibiotics in constructed wetland treatment systems based on key design and operational parameters: a review. *J Hazard Mater*. 407:124386. doi: [10.1016/j.jhazmat.2020.124386](https://doi.org/10.1016/j.jhazmat.2020.124386).
- Hou D, Sun Y, Wu X, Liu G, Zhu Y. 2017. Distribution characteristics and potential functions of bacterial communities in constructed wetlands used for treating heavy metals and organic matter in river sediments. *Ecol Eng*. 102:10–19. doi: [10.1016/j.ecoleng.2017.01.002](https://doi.org/10.1016/j.ecoleng.2017.01.002).
- Jiang Y, Xu C, Wu X, Chen Y, Han W, Gin KY, He Y. 2018. Occurrence, seasonal variation and risk assessment of antibiotics in Qingcaosha reservoir. *Water*. 10(2):115. doi: [10.3390/w10020115](https://doi.org/10.3390/w10020115).
- Kadlec RH, Wallace S. 2008. *Treatment wetlands* (2nd ed.). Florida, USA: CRC Press.
- Kelty CA, Varma M, Sivaganesan M, Haugland RA, Shanks OC. 2012. Distribution of genetic marker concentrations for fecal indicator bacteria in sewage and animal feces. *Appl Environ Microbiol*. 78(12):4225–4232. doi: [10.1128/AEM.06331-11](https://doi.org/10.1128/AEM.06331-11).
- Khawwaja NR, Polprasert C. 1999. Kinetics of fecal coliform removal in constructed wetlands. *WatSci Technol*. 40(3):109–116. doi: [10.2166/wst.1999.0147](https://doi.org/10.2166/wst.1999.0147).
- Kim I, Tanaka H. 2010. Photodegradation characteristics of PPCPs in water with UV treatment. *Environ Int*. 36(3):288–294. doi: [10.1016/j.envint.2009.11.004](https://doi.org/10.1016/j.envint.2009.11.004).
- Kümmerer K. 2009. Antibiotics in the environment: the role of wastewater treatment plants. *Environ Int*. 35(6):1005–1010. doi: [10.1016/j.envint.2009.04.003](https://doi.org/10.1016/j.envint.2009.04.003).
- Luo W, Li Y, Zhang L, Lin H, He Y. 2010. Hybrid constructed wetland systems for removing antibiotic-resistant bacteria and genes in wastewater treatment. *Environ Sci Technol*. 44(13):5218–5223. doi: [10.1021/es903514j](https://doi.org/10.1021/es903514j).
- Ma L, Li B, Jiang X, Wang Y, Xia Y, Li A. 2020. Removal of antibiotic-resistant bacteria and antibiotic resistance genes in wastewater effluent by constructed wetlands: effects of flow configuration and substrate type. *J Hazard Mater*. 381:120979. doi: [10.1016/j.jhazmat.2019.120979](https://doi.org/10.1016/j.jhazmat.2019.120979).
- Martínez-Martínez JG, Rosales-Loredo S, Hernández-Morales A, Arvizu-Gómez JL, Carranza-Álvarez C, Macías-Pérez JR, Rolón-Cárdenas GA, Pacheco-Aguilar JR. 2023. Bacterial communities associated with the roots of typha spp. and its relationship in phytoremediation processes. *Microorganisms*. 11(6):1587. doi: [10.3390/microorganisms11061587](https://doi.org/10.3390/microorganisms11061587).
- Matamoros V, Arias C, Brix H, Bayona JM. 2007. Removal of pharmaceuticals and personal care products (PPCPs) from urban wastewater in a pilot vertical flow constructed wetland and a sand filter. *Environ Sci Technol*. 41(23):8171–8177. doi: [10.1021/es0700707](https://doi.org/10.1021/es0700707).
- Morató J, Codony F, Sánchez O, Pérez LM, García J, Mas J. 2014. Key design factors affecting microbial community composition and pathogenic organism removal in horizontal subsurface flow constructed wetlands. *Sci Total Environ*. 481:81–89. doi: [10.1016/j.scitotenv.2014.01.068](https://doi.org/10.1016/j.scitotenv.2014.01.068).
- Paraskevopoulos S, Smeets P. 2019. Quantifying the log reduction of pathogenic microorganisms by constructed wetlands: a review. *Proceedings*. 48(1) 1–21. doi: [10.3390/ecws-4-06433](https://doi.org/10.3390/ecws-4-06433).
- Rizzo L, Manaia C, Merlin C, Schwartz T, Dagot C, Ploy MC, Michael I, Fatta-Kassinos D. 2013. Urban wastewater treatment plants as hotspots for antibiotic-resistant bacteria and genes spread into the environment: a review. *Sci Total Environ*. 447:345–360. doi: [10.1016/j.scitotenv.2013.01.032](https://doi.org/10.1016/j.scitotenv.2013.01.032).
- Rodríguez-Mozaz S, Vaz-Moreira I, Della Giustina SV, Llorca M, Barceló D, Schubert S, Berendonk TU, Michael-Kordatou I. 2020. Antibiotics in urban wastewater and rivers of the Iberian Peninsula: occurrence and removal. *Sci Total Environ*. 703:135550. doi: [10.1016/j.scitotenv.2019.135550](https://doi.org/10.1016/j.scitotenv.2019.135550).
- Rodríguez-Mozaz S, Vaz-Moreira I, Schubert S, Barceló D. 2015. Antibiotic resistance genes in wastewater treatment plants and their receiving environments: a review. *Water Res*. 67:150–170. doi: [10.1016/j.watres.2014.08.032](https://doi.org/10.1016/j.watres.2014.08.032).
- Sabri N, Schmitt H, Van Der Zaan B, Gerritsen H, Rijnaarts H, Langenhoff A. 2021. Performance of full scale constructed wetlands in removing antibiotics and antibiotic resistance genes. *Sci Total Environ*. 786:147368. doi: [10.1016/j.scitotenv.2021.147368](https://doi.org/10.1016/j.scitotenv.2021.147368).
- Saeed T, Alam MK, Miah MJ, Majed N. 2021. Removal of heavy metals in subsurface flow constructed wetlands: application of effluent recirculation. *Environ Sustain Indic*. 12:100146. doi: [10.1016/j.indic.2021.100146](https://doi.org/10.1016/j.indic.2021.100146).
- Santos FC, Almeida CMR, Ribeiro I, Mucha AP. 2019. Potential of constructed wetland for the removal of antibiotics and antibiotic resistant bacteria from livestock wastewater. *Ecol Eng*. 129:45–53. doi: [10.1016/j.ecoleng.2019.01.007](https://doi.org/10.1016/j.ecoleng.2019.01.007).
- Sehar S, Aamir R, Naz I, Ali N, Ahmed S. 2013. Reduction of contaminants (physical, chemical, and microbial) in domestic wastewater through hybrid constructed wetland. *ISRN Microbiol*. 2013:350260–350269. doi: [10.1155/2013/350260](https://doi.org/10.1155/2013/350260).
- Shafiquzzaman M, Haider H, Bhuiyan MA, Ahmed AT, AlSaleem SS, Ghumman AR. 2020. Spatiotemporal variations of DOM compo-

- nents in the Kushiro River impacted by a wetland. *Environ Sci Pollut Res Int.* 27(15):18287–18302. doi: [10.1007/s11356-020-08192-7](https://doi.org/10.1007/s11356-020-08192-7).
- Sudarsan JS, Annadurai R, Mukhopadhyay M, Chakraborty P, Nithiyantham S. 2018. Domestic wastewater treatment using constructed wetland: an efficient and alternative way. *Sustain Water Resour Manag.* 4(4):781–787. doi: [10.1007/s40899-017-0164-x](https://doi.org/10.1007/s40899-017-0164-x).
- Topal M, Arslan Topal EI. 2015. Occurrence and fate of tetracycline and degradation products in municipal biological wastewater treatment plant and transport of them in surface water. *Environ Monit Assess.* 187(12):750. doi: [10.1007/s10661-015-4956-x](https://doi.org/10.1007/s10661-015-4956-x).
- Topal M, Arslan Topal EI. 2020. Phytoremediation of priority substances (Pb and Ni) by *Phragmites australis*. *Int J Phytoremediation.* 22(8):857–862. doi: [10.1080/15226514.2020.1774504](https://doi.org/10.1080/15226514.2020.1774504).
- Topal M, Uslu Şenel G, Öbek E, Arslan Topal EI. 2015. Removal of tetracycline and the degradation products by *Lemna gibba* L. exposed to secondary effluents. *Env Prog and Sustain Energy.* 34(5):1311–1321. doi: [10.1002/ep.12125](https://doi.org/10.1002/ep.12125).
- Uluseker C, Kaster KM, Thorsen K, Basiry D, Shobana S, Jain M, Kumar G, Kommedal R, Pala-Ozkok I. 2021. A review on occurrence and spread of antibiotic resistance in wastewaters and in wastewater treatment plants: mechanisms and perspectives. *Front Microbiol.* 12:717809. doi: [10.3389/fmicb.2021.717809](https://doi.org/10.3389/fmicb.2021.717809).
- Vuotto C, Longo F, Balice MP, Donelli G, Varaldo PE. 2014. Antibiotic resistance related to biofilm formation in *Klebsiella pneumoniae*. *Pathogens.* 3(3):743–758. doi: [10.3390/pathogens3030743](https://doi.org/10.3390/pathogens3030743).
- Wang X, Hu M, Cheng F, Zhao S, Zhu D. 2020. Removal of antibiotic resistance genes in hybrid constructed wetlands. *Water Res.* 187:116136. doi: [10.1016/j.watres.2020.116136](https://doi.org/10.1016/j.watres.2020.116136).
- Yang S, Xu Y, Tang Z, Jin S, Yang S. 2024. The impact of alkaline stress on plant growth and its alkaline resistance mechanisms. *Int J Mol Sci.* 25(24):13719. doi: [10.3390/ijms252413719](https://doi.org/10.3390/ijms252413719).
- Yuan Q, Guo M, Yang J. 2015. Fate of antibiotic resistant bacteria and genes during wastewater chlorination: implication for antibiotic resistance control. *PLoS One.* 10(3):e0119403. doi: [10.1371/journal.pone.0119403](https://doi.org/10.1371/journal.pone.0119403).
- Zafar R, Bashir S, Nabi D, Arshad M. 2021. Occurrence and quantification of prevalent antibiotics in wastewater samples from Rawalpindi and Islamabad, Pakistan. *Sci Total Environ.* 764:142596. doi: [10.1016/j.scitotenv.2020.142596](https://doi.org/10.1016/j.scitotenv.2020.142596).
- Zhang D, Gersberg RM, Ng WJ, Tan SK. 2014. Removal of pharmaceuticals and personal care products in aquatic plant-based systems: a review. *Environ Pollut.* 184:620–639. doi: [10.1016/j.envpol.2013.09.002](https://doi.org/10.1016/j.envpol.2013.09.002).
- Zhang H, Chen Z, Chen J, Li J, Pan X. 2020. Influence of hydraulic retention time on the removal of ARGs in vertical subsurface flow constructed wetlands. *Environ Pollut.* 262: 114316. doi: [10.1016/j.envpol.2020.114316](https://doi.org/10.1016/j.envpol.2020.114316).
- Zhao X, Zhang Y, Liu Y, Wang H, Li P. 2023. Continuous inactivation of human adenoviruses in water by a novel g-C₃N₄/WO₃/biochar memory photocatalyst under light-dark cycles. *J Hazard Mater.* 442:130013. doi: [10.1016/j.jhazmat.2023.130013](https://doi.org/10.1016/j.jhazmat.2023.130013).